

AIM: Implementing Substitution and Transposition Ciphers

1A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques.

1B Design and implement algorithms to encrypt and decrypt messages using transposition techniques.

1A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques.

CODE:

plain = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"

cipher = "QWERTYUIOPASDFGHJKLZXCVBNM"

#Caesar Cipher

```
def caesar_encrypt(text, shift):
```

```
    result = ""
```

```
    for ch in text:
```

```
        if ch.isalpha():
```

```
            base = ord('A') if ch.isupper() else ord('a')
```

```
            result += chr((ord(ch) - base + shift) % 26 + base)
```

```
        else:
```

```
            result += ch
```

```
    return result
```

```
def caesar_decrypt(text, shift):
```

```
    return caesar_encrypt(text, -shift)
```

#Monoalphabetic Cipher

```
def mono_encrypt(text):
```

```
    result = ""
```

```
    for ch in text.upper():
```

```
        if ch.isalpha():
```

```
            index = plain.index(ch)
```

```
            result += cipher[index]
```

```
        else:
```

```
            result += ch
```

```
    return result
```

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```
def mono_decrypt(text):  
    result = ""  
    for ch in text.upper():  
        if ch.isalpha():  
            index = cipher.index(ch)  
            result += plain[index]  
        else:  
            result += ch  
    return result
```

#Main Program

while True:

```
    print("\n==== CLASSICAL SUBSTITUTION CIPHERS =====")  
    print("1. Caesar Cipher")  
    print("2. Monoalphabetic Cipher")  
    print("3. Exit")
```

```
    choice = input("Choose Technique: ")
```

```
    if choice == "1":
```

```
        print("\n1. Encrypt")  
        print("2. Decrypt")
```

```
        op = input("Choose Option: ")  
        msg = input("Enter Message: ")  
        shift = int(input("Enter Shift Value: "))
```

```
        if op == "1":
```

```
            print("Encrypted Message :", caesar_encrypt(msg, shift))
```

```
        elif op == "2":
```

```
            print("Decrypted Message :", caesar_decrypt(msg, shift))
```

```
        else:
```

```
            print("Invalid Option!")
```

```
    elif choice == "2":
```

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```
print("\n1. Encrypt")
print("2. Decrypt")

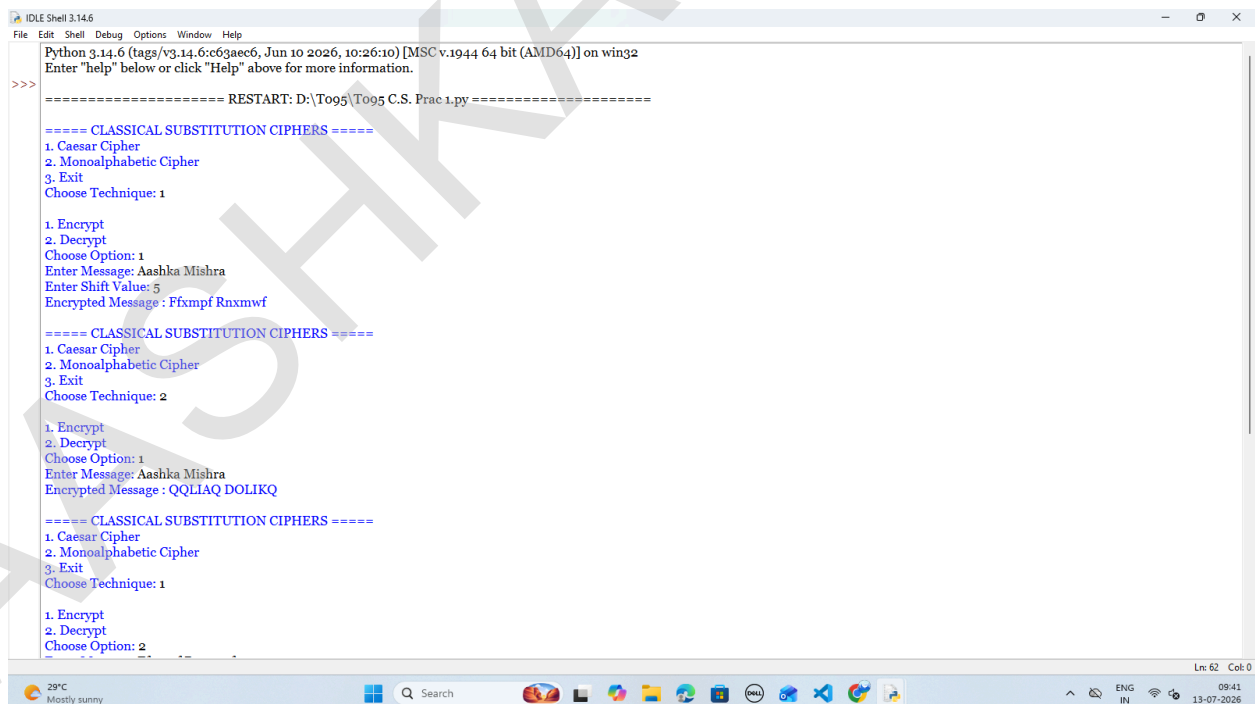
op = input("Choose Option: ")
msg = input("Enter Message: ")

if op == "1":
    print("Encrypted Message :", mono_encrypt(msg))
elif op == "2":
    print("Decrypted Message :", mono_decrypt(msg))
else:
    print("Invalid Option!")

elif choice == "3":
    print("Thank You!")
    break

else:
    print("Invalid Choice!")
```

OUTPUT:



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: D:\To95\To95 C.S. Prac 1.py =====

===== CLASSICAL SUBSTITUTION CIPHERS =====
1. Caesar Cipher
2. Monoalphabetic Cipher
3. Exit
Choose Technique: 1

1. Encrypt
2. Decrypt
Choose Option: 1
Enter Message: Aashka Mishra
Enter Shift Value: 5
Encrypted Message : Ffxmpf Rnxmwf

===== CLASSICAL SUBSTITUTION CIPHERS =====
1. Caesar Cipher
2. Monoalphabetic Cipher
3. Exit
Choose Technique: 2

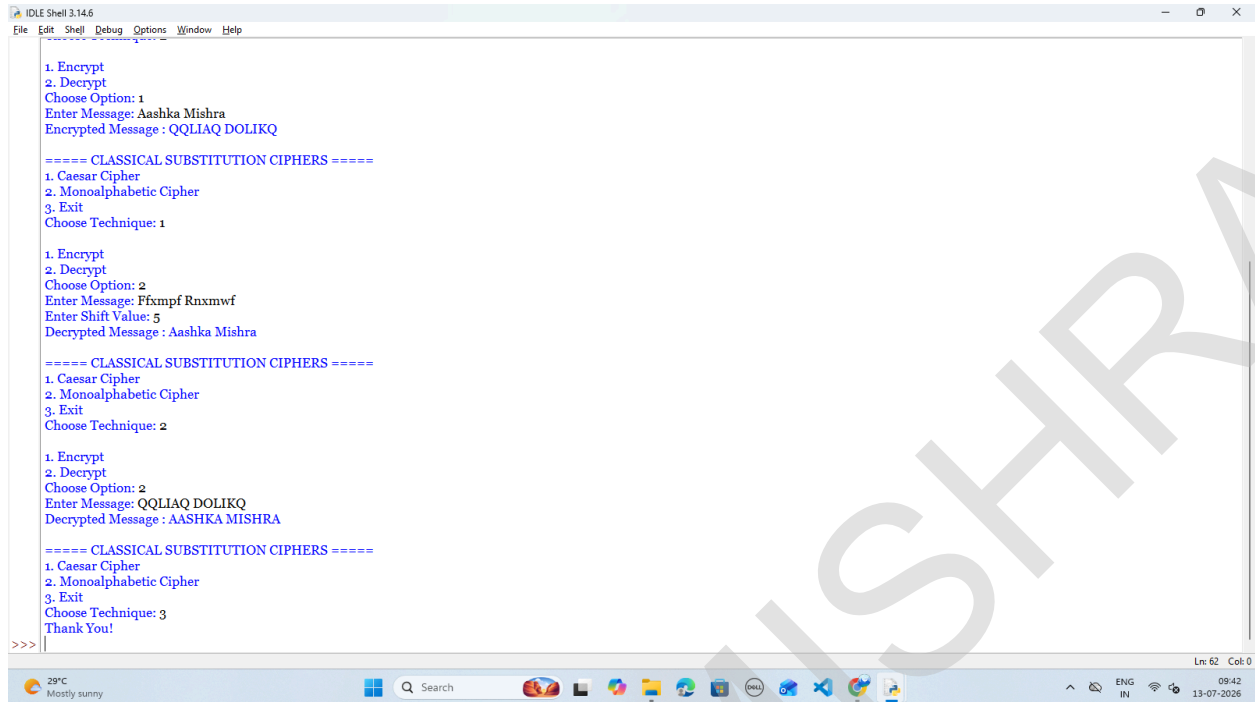
1. Encrypt
2. Decrypt
Choose Option: 1
Enter Message: Aashka Mishra
Encrypted Message : QQLIAQ DOLIKQ

===== CLASSICAL SUBSTITUTION CIPHERS =====
1. Caesar Cipher
2. Monoalphabetic Cipher
3. Exit
Choose Technique: 1

1. Encrypt
2. Decrypt
Choose Option: 2
```

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```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

1. Encrypt
2. Decrypt
Choose Option: 1
Enter Message: Aashka Mishra
Encrypted Message : QQLIAQ DOLIKQ

===== CLASSICAL SUBSTITUTION CIPHERS =====
1. Caesar Cipher
2. Monoalphabetic Cipher
3. Exit
Choose Technique: 1

1. Encrypt
2. Decrypt
Choose Option: 2
Enter Message: Ffxmpf Rnxmwf
Enter Shift Value: 5
Decrypted Message : Aashka Mishra

===== CLASSICAL SUBSTITUTION CIPHERS =====
1. Caesar Cipher
2. Monoalphabetic Cipher
3. Exit
Choose Technique: 2

1. Encrypt
2. Decrypt
Choose Option: 2
Enter Message: QQLIAQ DOLIKQ
Decrypted Message : AASHKA MISHRA

===== CLASSICAL SUBSTITUTION CIPHERS =====
1. Caesar Cipher
2. Monoalphabetic Cipher
3. Exit
Choose Technique: 3
Thank You!
>>>
```

GUI:

```
import tkinter as tk
```

```
from tkinter import ttk, messagebox
```

```
#Monoalphabetic Key
```

```
plain = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
```

```
cipher = "QWERTYUIOPASDFGHJKLZXCVBNM"
```

```
#Caesar Cipher
```

```
def caesar_encrypt(text, shift):
```

```
    result = ""
```

```
    for ch in text:
```

```
        if ch.isalpha():
```

```
            base = ord('A') if ch.isupper() else ord('a')
```

```
            result += chr((ord(ch) - base + shift) % 26 + base)
```

```
        else:
```

```
            result += ch
```

```
    return result
```

```
def caesar_decrypt(text, shift):
```

```
    return caesar_encrypt(text, -shift)
```

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#Monoalphabetic Cipher

```
def mono_encrypt(text):  
    result = ""  
    for ch in text.upper():  
        if ch.isalpha():  
            result += cipher[plain.index(ch)]  
        else:  
            result += ch  
    return result
```

```
def mono_decrypt(text):  
    result = ""  
    for ch in text.upper():  
        if ch.isalpha():  
            result += plain[cipher.index(ch)]  
        else:  
            result += ch  
    return result
```

#Functions

```
def encrypt():  
    text = message_entry.get()  
  
    if text == "":  
        messagebox.showwarning("Warning", "Please enter a message.")  
        return  
  
    if combo.get() == "Caesar Cipher":  
        try:  
            shift = int(shift_entry.get())  
            result.set(caesar_encrypt(text, shift))  
        except:  
            messagebox.showerror("Error", "Enter a valid shift value.")  
    else:  
        result.set(mono_encrypt(text))  
  
def decrypt():  
    text = message_entry.get()
```

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```
if text == "":
    messagebox.showwarning("Warning", "Please enter a message.")
    return

if combo.get() == "Caesar Cipher":
    try:
        shift = int(shift_entry.get())
        result.set(caesar_decrypt(text, shift))
    except:
        messagebox.showerror("Error", "Enter a valid shift value.")
    else:
        result.set(mono_decrypt(text))

def clear():
    message_entry.delete(0, tk.END)
    shift_entry.delete(0, tk.END)
    result.set("")

#Main Window
root = tk.Tk()
root.title("Classical Substitution Cipher")
root.geometry("650x500")
root.configure(bg="#0F172A")
root.resizable(False, False)

#Heading
title = tk.Label(
    root,
    text="🔒 Classical Substitution Cipher System",
    font=("Segoe UI", 20, "bold"),
    bg="#1E3A8A",
    fg="white",
    pady=15
)
title.pack(fill="x")

#Card
card = tk.Frame(root, bg="white", bd=2, relief="ridge")
```

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```
card.place(relx=0.5, rely=0.52, anchor="center", width=560, height=360)
```

```
#Labels
```

```
tk.Label(card,  
    text="Cipher Technique",  
    font=("Segoe UI", 11, "bold"),  
    bg="white").place(x=40, y=30)
```

```
combo = ttk.Combobox(  
    card,  
    values=["Caesar Cipher", "Monoalphabetic Cipher"],  
    state="readonly",  
    width=28,  
    font=("Segoe UI", 10)  
)  
combo.current(0)  
combo.place(x=220, y=30)
```

```
tk.Label(card,  
    text="Enter Message",  
    font=("Segoe UI", 11, "bold"),  
    bg="white").place(x=40, y=80)
```

```
message_entry = tk.Entry(  
    card,  
    font=("Segoe UI", 11),  
    width=32,  
    relief="solid",  
    bd=1  
)  
message_entry.place(x=220, y=82)
```

```
tk.Label(card,  
    text="Shift Value",  
    font=("Segoe UI", 11, "bold"),  
    bg="white").place(x=40, y=130)
```

```
shift_entry = tk.Entry(  
    card,  
    font=("Segoe UI", 11),
```

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```
width=10,  
relief="solid",  
bd=1  
)  
shift_entry.place(x=220, y=132)  
  
#Buttons  
encrypt_btn = tk.Button(  
    card,  
    text="🔒 Encrypt",  
    font=("Segoe UI", 11, "bold"),  
    bg="#16A34A",  
    fg="white",  
    width=12,  
    command=encrypt,  
    cursor="hand2"  
)  
encrypt_btn.place(x=55, y=190)  
  
decrypt_btn = tk.Button(  
    card,  
    text="🔓 Decrypt",  
    font=("Segoe UI", 11, "bold"),  
    bg="#2563EB",  
    fg="white",  
    width=12,  
    command=decrypt,  
    cursor="hand2"  
)  
decrypt_btn.place(x=215, y=190)  
  
clear_btn = tk.Button(  
    card,  
    text="🗑️ Clear",  
    font=("Segoe UI", 11, "bold"),  
    bg="#DC2626",  
    fg="white",  
    width=12,  
    command=clear,  
    cursor="hand2"
```


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```
)
clear_btn.place(x=375, y=190)

#Result
tk.Label(card,
          text="Result",
          font=("Segoe UI", 11, "bold"),
          bg="white").place(x=40, y=265)

result = tk.StringVar()

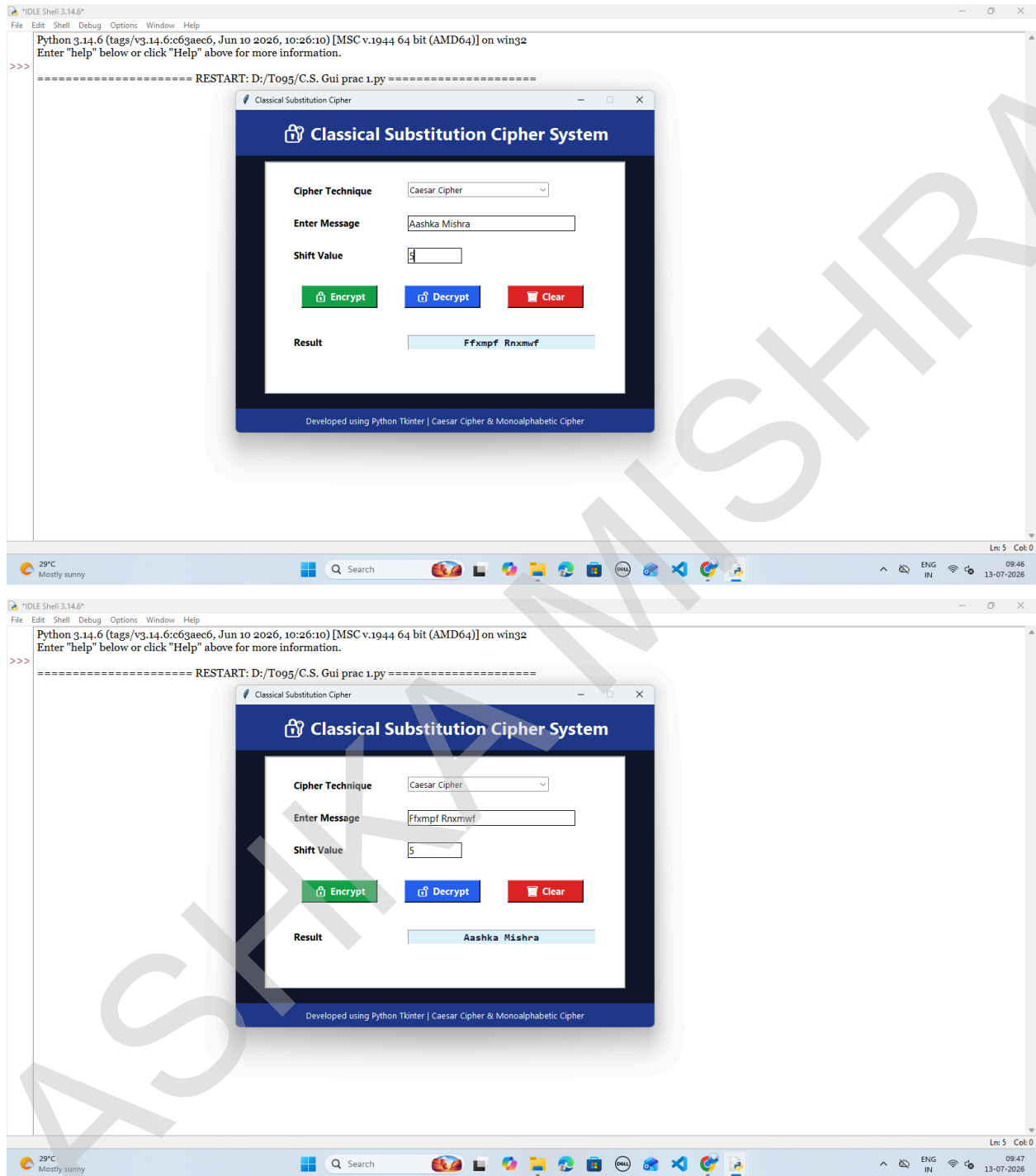
result_box = tk.Entry(
    card,
    textvariable=result,
    font=("Consolas", 12, "bold"),
    width=32,
    justify="center",
    state="readonly",
    readonlybackground="#E0F2FE",
    fg="#0F172A"
)
result_box.place(x=220, y=267)

#Footer
footer = tk.Label(
    root,
    text="Developed using Python Tkinter | Caesar Cipher & Monoalphabetic Cipher",
    font=("Segoe UI", 10),
    bg="#1E3A8A",
    fg="white",
    pady=8
)
footer.pack(side="bottom", fill="x")

root.mainloop()
```

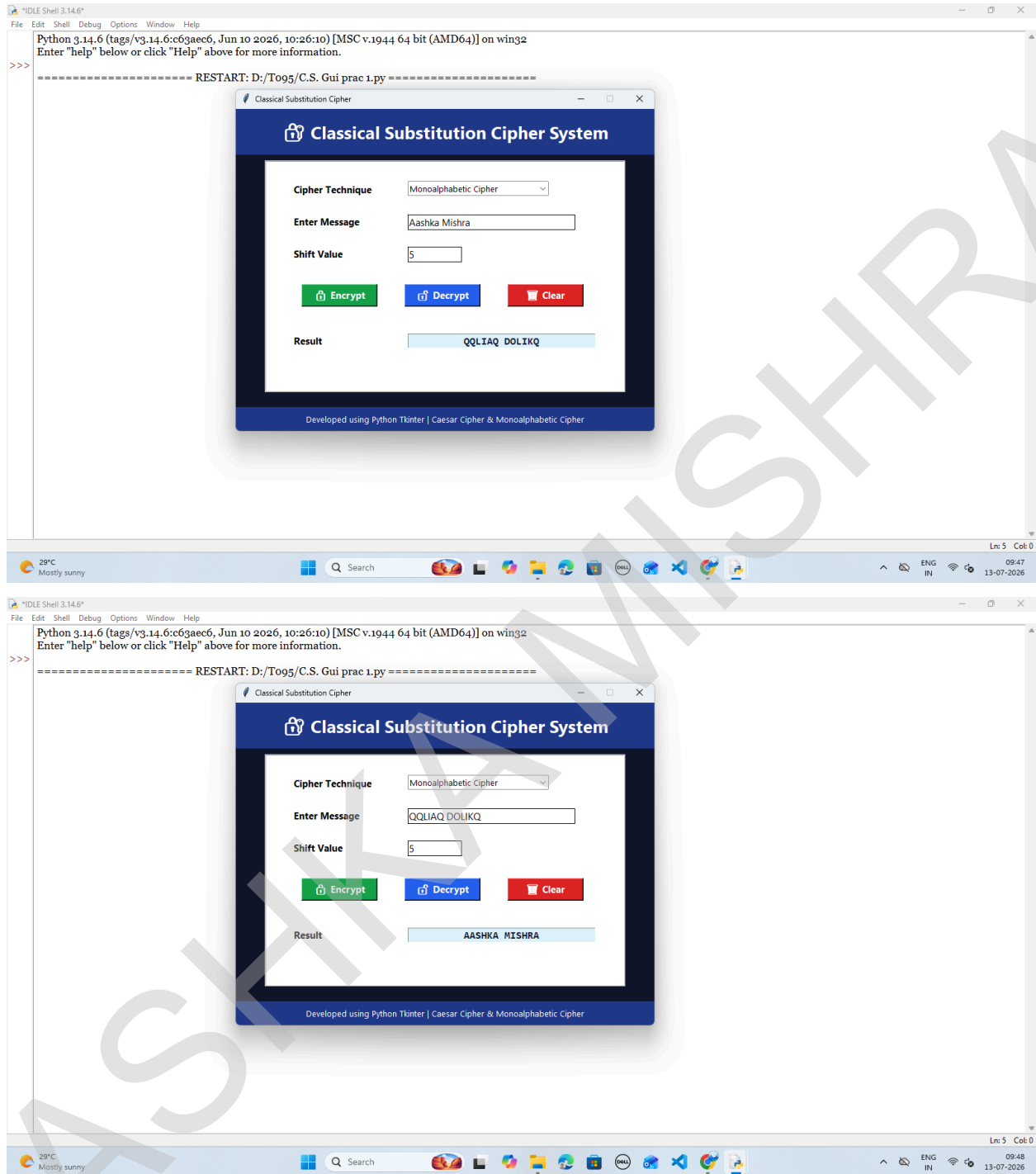
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OUTPUT:



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1B Design and implement algorithms to encrypt and decrypt messages using transposition techniques.

CODE:

#Rail Fence Cipher

```
def rail_encrypt(text, key):
```

```
    rail = [""] * key
```

```
    row, step = 0, 1
```

```
    for ch in text:
```

```
        rail[row] += ch
```

```
        if row == 0:
```

```
            step = 1
```

```
        elif row == key - 1:
```

```
            step = -1
```

```
        row += step
```

```
    return "".join(rail)
```

```
def rail_decrypt(cipher, key):
```

```
    n = len(cipher)
```

```
    mark = [[0] * n for _ in range(key)]
```

```
    row, step = 0, 1
```

```
    for i in range(n):
```

```
        mark[row][i] = 1
```

```
        if row == 0:
```

```
            step = 1
```

```
        elif row == key - 1:
```

```
            step = -1
```

```
        row += step
```

```
    k = 0
```

```
    for i in range(key):
```

```
        for j in range(n):
```

```
            if mark[i][j]:
```

```
                mark[i][j] = cipher[k]
```

```
                k += 1
```

```
    row, step = 0, 1
```

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```
text = ""
for i in range(n):
    text += mark[row][i]
    if row == 0:
        step = 1
    elif row == key - 1:
        step = -1
    row += step

return text
```

#Columnar Transposition

```
def col_encrypt(text, key):
    cols = len(key)
    rows = (len(text) + cols - 1) // cols

    text += "X" * (rows * cols - len(text))

    matrix = [list(text[i:i + cols]) for i in range(0, len(text), cols)]
    order = sorted(range(cols), key=lambda i: key[i])

    cipher = ""
    for c in order:
        for r in matrix:
            cipher += r[c]
    return cipher

def col_decrypt(cipher, key):
    cols = len(key)
    rows = len(cipher) // cols

    order = sorted(range(cols), key=lambda i: key[i])

    matrix = ["" * cols for _ in range(rows)]

    k = 0
    for c in order:
        for r in range(rows):
```

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```
matrix[r][c] = cipher[k]  
k += 1
```

```
text = ""  
for r in matrix:  
    text += "".join(r)  
  
return text.rstrip("X")
```

#Main Menu

while True:

```
print("\n===== TRANSPOSITION CIPHERS =====")  
print("1. Rail Fence Cipher")  
print("2. Columnar Transposition")  
print("3. Exit")
```

```
choice = input("Enter choice: ")
```

```
if choice == "3":  
    print("Thank You!")  
    break
```

```
print("\n1. Encrypt")  
print("2. Decrypt")  
op = input("Enter option: ")
```

```
msg = input("Enter Message: ")
```

```
if choice == "1":  
    key = int(input("Enter Rail Key: "))
```

```
if op == "1":  
    print("Encrypted:", rail_encrypt(msg, key))  
elif op == "2":  
    print("Decrypted:", rail_decrypt(msg, key))  
else:  
    print("Invalid Option!")
```

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```
elif choice == "2":  
    key = input("Enter Key (Example: ZEBRA): ").upper()  
  
    if op == "1":  
        print("Encrypted:", col_encrypt(msg, key))  
    elif op == "2":  
        print("Decrypted:", col_decrypt(msg, key))  
    else:  
        print("Invalid Option!")  
  
else:  
    print("Invalid Choice!")
```

OUTPUT:

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```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:6c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

===== RESTART: D:/T095/T095 C.S. Prac 1 B.py =====

===== TRANSPOSITION CIPHERS =====
1. Rail Fence Cipher
2. Columnar Transposition
3. Exit
Enter choice: 1

1. Encrypt
2. Decrypt
Enter option: 1
Enter Message: AASHKA MISHRA
Enter Rail Key: 2
Encrypted: ASK IHAAHAMSR

===== TRANSPOSITION CIPHERS =====
1. Rail Fence Cipher
2. Columnar Transposition
3. Exit
Enter choice: 1

1. Encrypt
2. Decrypt
Enter option: 2
Enter Message: ASK IHAAHAMSR
Enter Rail Key: 2
Decrypted: AASHKA MISHRA

===== TRANSPOSITION CIPHERS =====
1. Rail Fence Cipher
2. Columnar Transposition
3. Exit
Enter choice: 2

1. Encrypt
2. Decrypt
Enter option: 2
Enter Message: ASK IHAAHAMSR
Enter Rail Key: 2
Decrypted: AASHKA MISHRA

===== TRANSPOSITION CIPHERS =====
1. Rail Fence Cipher
2. Columnar Transposition
3. Exit
Enter choice: 2

1. Encrypt
2. Decrypt
Enter option: 1
Enter Message: aashka mishra
Enter Key (Example: ZEBRA): panda
Encrypted: a rksXhiXsmaaah

===== TRANSPOSITION CIPHERS =====
1. Rail Fence Cipher
2. Columnar Transposition
3. Exit
Enter choice: 2

1. Encrypt
2. Decrypt
Enter option: 2
Enter Message: a rksXhiXsmaaah
Enter Key (Example: ZEBRA): panda
Decrypted: aashka mishra

===== TRANSPOSITION CIPHERS =====
1. Rail Fence Cipher
2. Columnar Transposition
3. Exit
Enter choice: 3
Thank You!
>>>
```

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GUI:

```
import tkinter as tk
from tkinter import ttk, messagebox
```

#Rail Fence Cipher

```
def rail_encrypt(text, key):
```

```
    rail = [""] * key
    row, step = 0, 1
    for ch in text:
        rail[row] += ch
        if row == 0:
            step = 1
        elif row == key - 1:
            step = -1
        row += step
    return "".join(rail)
```

```
def rail_decrypt(cipher, key):
```

```
    n = len(cipher)
    mark = [[0] * n for _ in range(key)]
```

```
    row, step = 0, 1
    for i in range(n):
        mark[row][i] = 1
        if row == 0:
            step = 1
        elif row == key - 1:
            step = -1
        row += step
```

```
    k = 0
    for i in range(key):
        for j in range(n):
            if mark[i][j]:
                mark[i][j] = cipher[k]
                k += 1
```

```
    row, step = 0, 1
    text = ""
    for i in range(n):
```

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```
text += mark[row][i]
if row == 0:
    step = 1
elif row == key - 1:
    step = -1
row += step
```

```
return text
```

#Columnar Cipher

```
def col_encrypt(text, key):
```

```
    cols = len(key)
    rows = (len(text) + cols - 1) // cols
```

```
    text += "X" * (rows * cols - len(text))
```

```
    matrix = [list(text[i:i+cols]) for i in range(0, len(text), cols)]
    order = sorted(range(cols), key=lambda i: key[i])
```

```
    cipher = ""
    for c in order:
        for r in matrix:
            cipher += r[c]
    return cipher
```

```
def col_decrypt(cipher, key):
```

```
    cols = len(key)
    rows = len(cipher) // cols
    order = sorted(range(cols), key=lambda i: key[i])
```

```
    matrix = ["" * cols for _ in range(rows)]
```

```
    k = 0
    for c in order:
        for r in range(rows):
            matrix[r][c] = cipher[k]
            k += 1
```

```
    text = ""
    for r in matrix:
```

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```
text += "".join(r)

return text.rstrip("X")

#Functions
def encrypt():
    msg = message.get()

    if msg == "":
        messagebox.showwarning("Warning", "Enter a message!")
        return

    try:
        if cipher_type.get() == "Rail Fence Cipher":
            key = int(key_entry.get())
            result.set(rail_encrypt(msg, key))
        else:
            key = key_entry.get().upper()
            result.set(col_encrypt(msg, key))
    except:
        messagebox.showerror("Error", "Invalid Key!")

def decrypt():
    msg = message.get()

    if msg == "":
        messagebox.showwarning("Warning", "Enter a message!")
        return

    try:
        if cipher_type.get() == "Rail Fence Cipher":
            key = int(key_entry.get())
            result.set(rail_decrypt(msg, key))
        else:
            key = key_entry.get().upper()
            result.set(col_decrypt(msg, key))
    except:
        messagebox.showerror("Error", "Invalid Key!")

def clear():
```

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```
message.delete(0, tk.END)
key_entry.delete(0, tk.END)
result.set("")
```

#GUI

```
root = tk.Tk()
root.title("Transposition Cipher")
root.geometry("650x500")
root.configure(bg="#0F172A")
root.resizable(False, False)
```

Header

```
tk.Label(
    root,
    text="🔒 Transposition Substitution Techniques",
    font=("Segoe UI", 20, "bold"),
    bg="#1E3A8A",
    fg="white",
    pady=15
).pack(fill="x")
```

Card

```
card = tk.Frame(root, bg="white", bd=2, relief="ridge")
card.place(relx=0.5, rely=0.52, anchor="center", width=560, height=360)
```

Technique

```
tk.Label(card, text="Technique", font=("Segoe UI", 11, "bold"),
    bg="white").place(x=40, y=30)
```

cipher_type = ttk.Combobox(

```
    card,
    values=["Rail Fence Cipher", "Columnar Transposition"],
    state="readonly",
    width=28,
    font=("Segoe UI", 10)
```

)

```
cipher_type.current(0)
cipher_type.place(x=220, y=30)
```

Message

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```
tk.Label(card, text="Message", font=("Segoe UI", 11, "bold"),  
        bg="white").place(x=40, y=80)
```

```
message = tk.Entry(card, width=32, font=("Segoe UI", 11))  
message.place(x=220, y=80)
```

Key

```
tk.Label(card, text="Key", font=("Segoe UI", 11, "bold"),  
        bg="white").place(x=40, y=130)
```

```
key_entry = tk.Entry(card, width=15, font=("Segoe UI", 11))  
key_entry.place(x=220, y=130)
```

Buttons

```
tk.Button(card, text="🔒 Encrypt",  
         bg="#16A34A", fg="white",  
         font=("Segoe UI", 11, "bold"),  
         width=12,  
         command=encrypt).place(x=45, y=190)
```

```
tk.Button(card, text="🔓 Decrypt",  
         bg="#2563EB", fg="white",  
         font=("Segoe UI", 11, "bold"),  
         width=12,  
         command=decrypt).place(x=210, y=190)
```

```
tk.Button(card, text="🗑️ Clear",  
         bg="#DC2626", fg="white",  
         font=("Segoe UI", 11, "bold"),  
         width=12,  
         command=clear).place(x=375, y=190)
```

Result

```
tk.Label(card, text="Result",  
        font=("Segoe UI", 11, "bold"),  
        bg="white").place(x=40, y=270)
```

```
result = tk.StringVar()
```

```
tk.Entry(card,
```

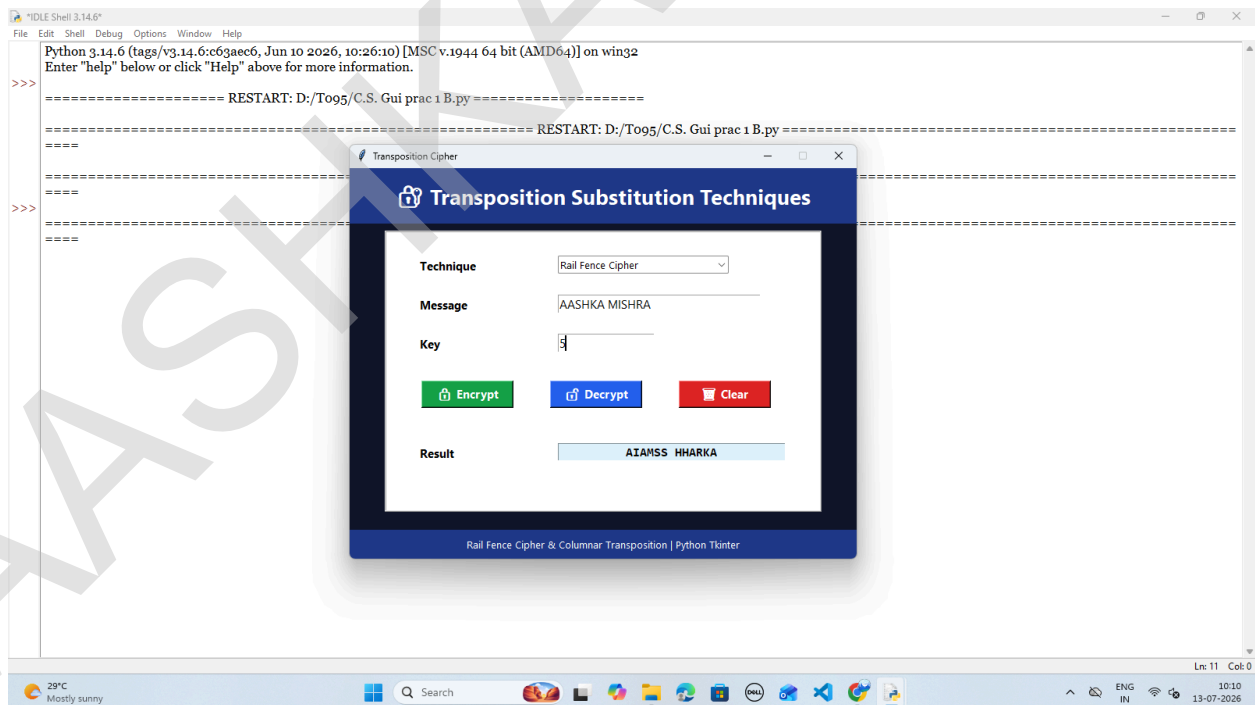
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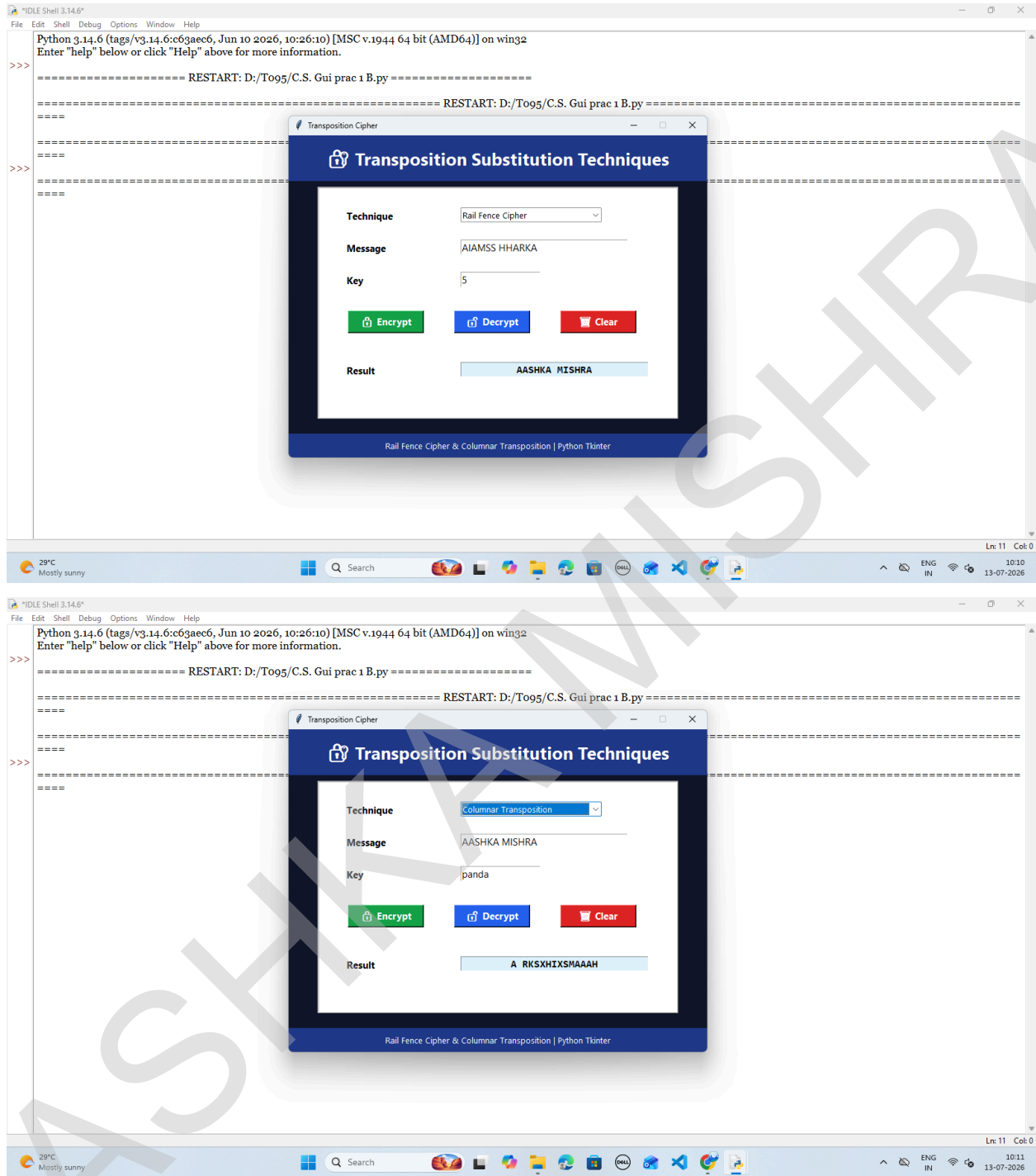
```
textvariable=result,  
font=("Consolas",12,"bold"),  
width=32,  
justify="center",  
state="readonly",  
readonlybackground="#E0F2FE").place(x=220, y=270)  
  
# Footer  
tk.Label(  
    root,  
    text="Rail Fence Cipher & Columnar Transposition | Python Tkinter",  
    bg="#1E3A8A",  
    fg="white",  
    font=("Segoe UI",10),  
    pady=8  
) .pack(side="bottom", fill="x")  
  
root.mainloop()
```

OUTPUT:



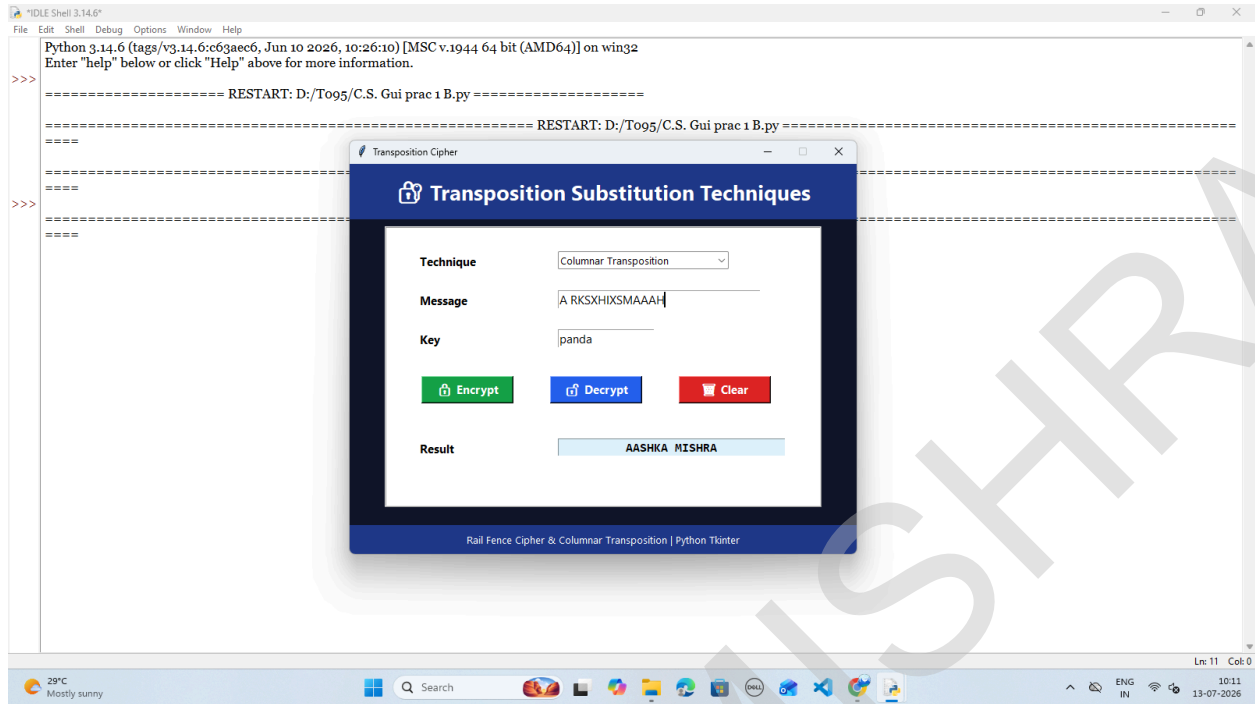
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